

How To Select The RIGHT TOUCHSCREEN PANEL



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Today there are almost 20 different technologies that revolve around touch panels. And there are millions of end products that use touch panels for navigation. We have come a long way in technology but how do these touch panels actually work? This aspect plays an important role in selection of such touchscreen panels.

EFY buyers' guide is intended to provide an overview of the present state of touchscreen panel technology and its selection criteria to allow selection of the most appropriate touch technology for the intended application.

Selecting the right touchscreen is critical to the product's success

A touchscreen is a human-machine interface. It is activated by the user by touching, with a stylus or single- or multi-finger touch, the selected area on the display that indicates the desired function. The touchscreen or frame contains an independent x-y coordinate system that is calibrated to the display matrix. The touch point is determined by the embedded computer by the x-y coordinates of the touch point. You, as a

user, without using a mouse or keyboard, can interact directly with the information displayed on the touchscreen.

Touch input can provide instant audio and/or visual feedback to you to confirm your choice. Touchscreens are common in devices such as smartphones, game consoles, personal computers, touchscreen TVs, iPads and electronic voting machines, among others. These are widely used in applications such as public information kiosks, industrial and process control, military displays and interactive video systems.

The touchscreen sensor and firmware for the host computer have been made available by a wide array of after-market system integrators, and not by display or chip manufacturers. Display manufacturers and chip manufacturers worldwide have accepted the trend toward acceptance of touchscreens as a highly-desirable user interface component and have begun to integrate touchscreens into the fundamental design of their products.

Ideally, a touchscreen would have the following characteristics:

- As it is layered on top of the display, it should not degrade the display image.
- Its resolution should be equal to that of the video display.
- Activation is independent of size of stylus, conducting or non-conducting nature of styli and touch of gloved or non-gloved finger.
- It is capable of providing positive, tactile feedback to the operator.
- Architecture should be modular and capable of integration with other input devices and networks.
- Due consideration has been given to the human perspective by considering aspects such as ergonomics, cognitive psychology and human physiology.
- It requires precise selection of targets to avoid accidental selection of adjacent targets.

